

Version 1.7 (Hackathon MVP)

Atlas

The Zero-Thought Study Operating System

Status: Approved for Development

Document Type: Product Requirements Document

Confidential & Proprietary

1. Executive Summary

Atlas is an AI-powered Study Operating System that eliminates the mental effort involved in planning and managing studies. Rather than asking students to organize subjects, create schedules, or decide what to revise, Atlas automatically generates personalized daily study missions based on their syllabus, exam dates, available study time, and previous progress.

Core Philosophy:

Students should never have to wonder what to study next. They should simply open Atlas, and the system already knows.

2. Problem Statement

Today's students rely on a highly fragmented ecosystem of applications to manage their academics. Each application solves one isolated problem, but none understands the student's complete academic journey:

- **ChatGPT:** Used for quick explanations.
- **Google Calendar:** Used for scheduling and deadlines.
- **Notion:** Used for planning and note-taking.
- **Forest:** Used for focus and time management.
- **Anki / Quizlet:** Used for flashcard revision.

As a result, students experience decision fatigue, poor prioritization, inconsistent revision cycles, forgotten concepts, and unstructured study sessions. The friction of context-switching means students often spend more time deciding *what* to study than actually studying.

3. Vision & Product Principles

Vision: Atlas should become the absolute first application a student opens before every study session. Instead of asking, "What should I study today?", students open Atlas, and the frictionless journey begins.

Core Principles

- **Zero-Thought Planning:** Students should never manually organize their studies. Atlas always calculates and provides the next best task.
- **Coach, Not Chatbot:** Atlas doesn't simply spoon-feed answers. It identifies misconceptions, recommends structural improvements, and guides cognitive learning.
- **Progress Over Hours:** True learning matters more than arbitrary study time. Atlas measures mastery and understanding rather than hours logged.
- **Momentum Over Streaks:** Missing one day should not psychologically penalize or erase weeks of progress. Momentum decreases gradually instead of rigidly resetting to zero.
- **Adapt Everything:** Every interaction, quiz result, and focus session feeds back into the engine to make better recommendations tomorrow.

4. Target Users

SEGMENT	DEMOGRAPHIC	KEY NEEDS
Primary	High school students (Ages 14–18), specifically targeting CBSE, ICSE, and State Boards.	Structure, board exam prep, continuous revision, stress reduction.
Secondary	University students & Competitive exam aspirants (JEE, NEET, SAT).	High-volume retention, deep analytics, spaced repetition.

5. Core Product Pillars

Atlas is built around five core experiences designed to encompass the entire learning lifecycle:



PLAN

Atlas automatically decides what the student should study based on variables like exam proximity and topic weaknesses. **Features:** AI Planner, Daily Missions, Interactive Calendar, Exam Timeline.



LEARN

A completely distraction-free, immersive study experience. **Features:** Focus Mode (Pomodoro), AI Coach, AI-Generated Quizzes, On-the-fly AI Explanations.



TRACK

Visualizing the invisible process of learning. **Features:** Dynamic Learning Graph, Topic Confidence Scoring, Granular Subject Progress.



ADAPT

An intelligent memory system that remembers previous sessions and predicts decay. **Features:** Confidence Decay Tracking, Automated Revision Queue, Deep Learning History.



IMPROVE

Gamification designed to motivate continuous, sustainable progress. **Features:** XP System, Momentum Physics, Mission Completion Streaks, Advanced Study Statistics.

6. User Journey Overview

The core Atlas loop is designed to minimize friction at every stage of the student's day.

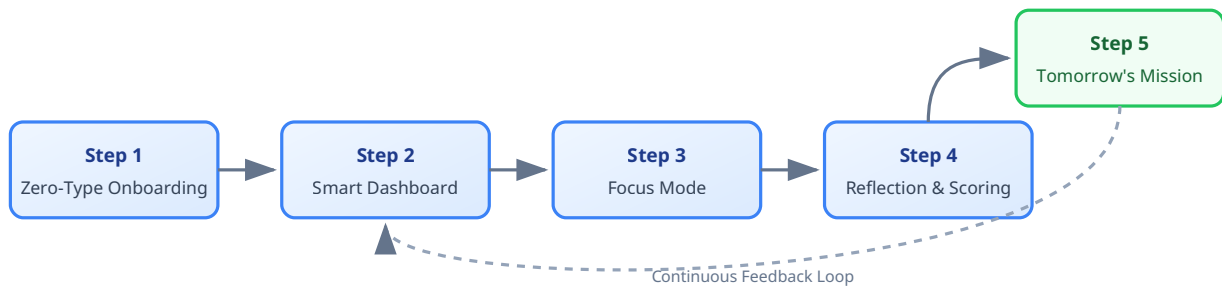


Figure 1: The Daily Atlas User Journey Loop

7. Core Features

7.1 Dashboard & AI Planner

The nerve center of Atlas. Instead of a blank slate, the user is greeted with **Today's Mission**. The AI Planner computes this mission by analyzing the user's available study time, upcoming exam dates, current topic confidence levels, and historical data from previous missions. It outputs a suggested revision order and estimated duration.

7.2 Interactive Calendar

A timeline of the academic journey. It visually displays upcoming exams alongside past completed daily missions and projected future revision sessions. This helps students grasp their long-term workload at a glance.

7.3 Focus Mode & AI Coach

Once a mission begins, the user enters Focus Mode. This includes a built-in Pomodoro timer to manage cognitive load. During study, the AI Coach provides Socratic-style hints, generates dynamic questions to test understanding, and performs misconception analysis to course-correct the student instantly.

7.4 The Learning Graph

A visual representation of the student's knowledge hierarchy. Selecting a node immediately initializes a micro-revision session.

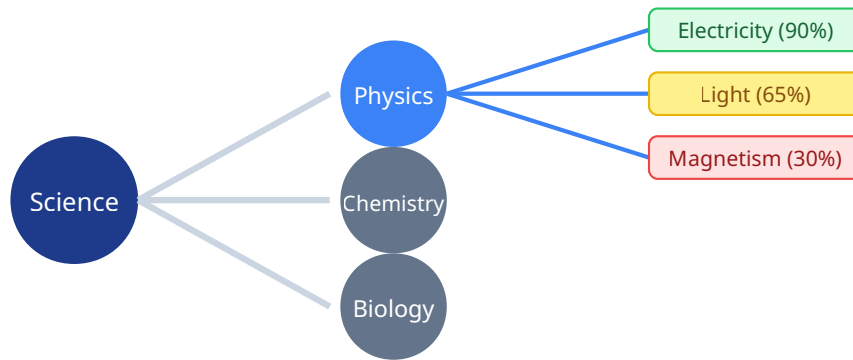


Figure 2: Conceptual rendering of the interactive Learning Graph (Physics Expanded)

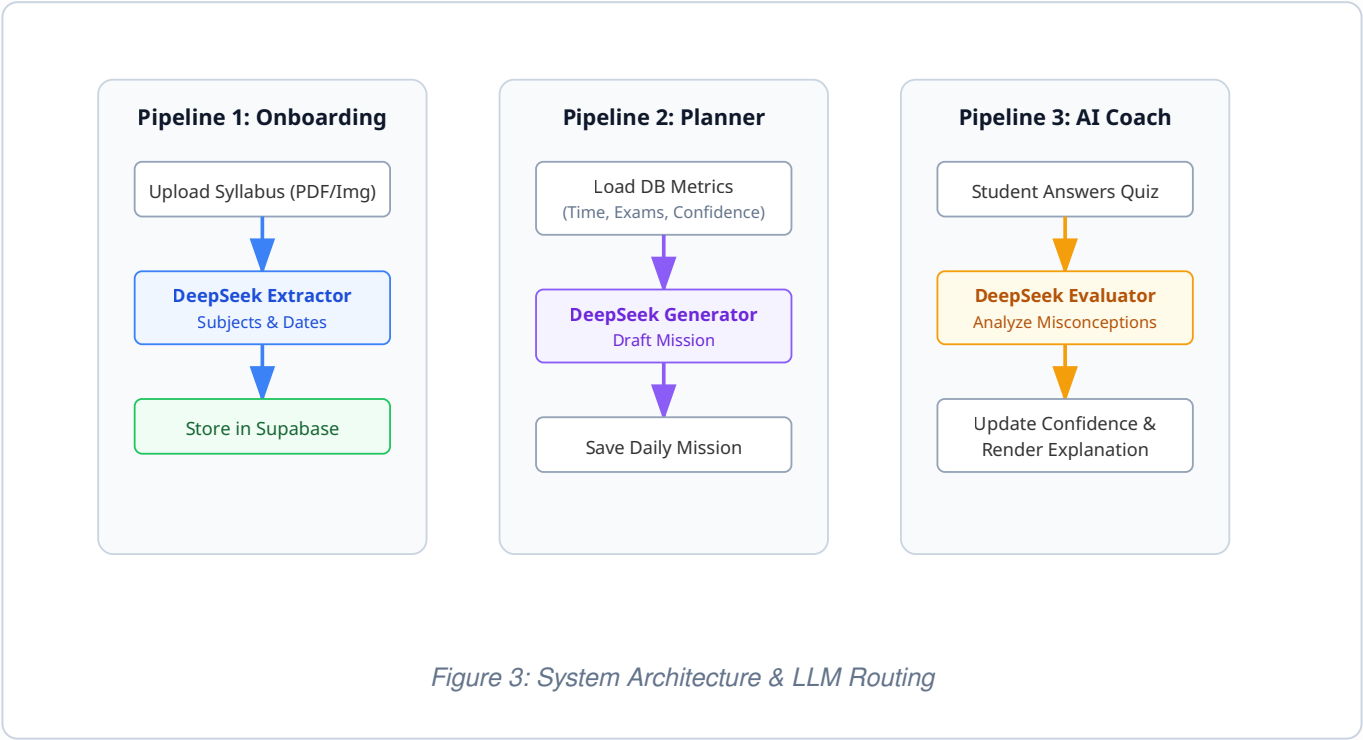
8. Database Schema (Supabase)

The backend requires a relational structure to link users, their subjects, specific topics, and dynamically generated daily missions.

TABLE	COLUMNS / SCHEMA DETAILS
users	id (UUID), study_time (Int), momentum (Float)
subjects	id (UUID), name (String), exam_date (Timestamp)
topics	id (UUID), subject_id (FK), confidence (Float, 0-100), next_review (Timestamp)
missions	id (UUID), user_id (FK), date (Date), status (Enum: Pending, Complete)
mission_tasks	id (UUID), mission_id (FK), topic_id (FK), status (Enum)

9. AI Pipelines Architecture

Atlas utilizes DeepSeek V4 Flash for ultra-fast, intelligent orchestration across three distinct pipelines.



10. Technical Stack & MVP Scope

LAYER	TECHNOLOGY
Frontend	Next.js (App Router), Tailwind CSS, Framer Motion
Backend	FastAPI
Database	Supabase (PostgreSQL)
Auth	Clerk
AI Engine	DeepSeek V4 Flash

LAYER	TECHNOLOGY
Deployment	Vercel (Frontend) + Railway (Backend)

MVP Scope (Included for Hackathon)

✔ Zero-Type Onboarding ✔ Dashboard ✔ AI Planner ✔ Calendar ✔ Focus Mode ✔ AI Coach ✔ Learning Graph ✔ Progress Stats (XP & Momentum)

Note on V2 Roadmap: Event-driven Atlas Brain, Elo Confidence Engine, Adaptive Spaced Repetition, and Mobile Apps are explicitly deferred to ensure a stable, demo-ready MVP.

11. Success Metrics (KPIs)

Atlas defines success through user consistency rather than raw time-in-app. Missing a day drops momentum, ensuring the student is gently incentivized to return.

- **Primary KPI:** Daily Mission Completion Rate (Target: 70%+)
- **Secondary KPIs:**
 - Daily Active Users (DAU)
 - Weekly Retention Rate
 - Average Mission Completion Time (to ensure AI estimates are accurate)
 - User Satisfaction (NPS)